

Notes on Probability

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4 Denumerable Probabilities

For a sequence of sets $\{A_n\}_n$, we define the sets:

$$\limsup_n A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k$$
$$\liminf_n A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k$$

which are the **limits superior** and **limits inferior** of the sequence A_n , respectively. We say that A_n has **limit** A (denoted $A_n \rightarrow A$) whenever:

$$\lim_n A_n = \liminf_n A_n = \limsup_n A_n$$

Note that $x \in \limsup_n A_n$ if and only if x lies in infinitely many of the A_n . Also, $x \in \liminf_n A_n$ is x lies in all but finitely many of the A_n . Therefore, if $x \in \liminf_n A_n$, then $x \in \limsup_n A_n$ since if it lies in all but finitely many of the A_n , it lies in infinitely many of the A_n . Hence, $\liminf_n A_n \subset \limsup_n A_n$. To check that $A_n \rightarrow A$, we only have to show $\limsup_n A_n \subset \liminf_n A_n$.

Theorem 4.1.

1. For each sequence $\{A_n\}_n$,

$$\begin{aligned} P\left(\liminf_n A_n\right) &\leq \inf_n P(A_n) \\ &\leq \limsup_n P(A_n) \leq P\left(\limsup_n A_n\right) \end{aligned}$$

2. If $A_n \rightarrow A$, then $P(A_n) \rightarrow P(A)$

Proof. We'll address each case separately.

- (i) (a) TODO
- (b) Put $l = \inf_n P(A_n)$. Then:

$$l \leq P(A_n) \leq \sup_{k \geq n} P(A_k)$$

for any particular choice of $n \in \mathbb{N}$. So, $l \leq \sup_{k \geq n} P(A_k)$ for all $n \in \mathbb{N}$. That is, it is a lower bound. However, $\inf_n \sup_{k \geq n} P(A_k)$ is the greatest lower bound for the sequence of $\left(\sup_{k \geq n} P(A_k)\right)_n$. So, $l \leq \inf_n \sup_{k \geq n} P(A_k)$. That is:

$$\inf_n P(A_n) \leq \inf_n \sup_{k \geq n} P(A_k) = \limsup_n P(A_n)$$

where we have used the results from Problem 2 for the definition of $\limsup_n P(A_n)$ for numerical sequences.

- (ii) From the preceding part, we know that $\lim_{n \rightarrow \infty} P(A_n) \leq P(\limsup_n A_n)$ and that $P(\liminf_n A_n) \leq \lim_{n \rightarrow \infty} P(A_n)$. If $A_n \rightarrow A$, then $\limsup_n A_n = \liminf_n A_n = A$, so $\lim_{n \rightarrow \infty} P(A_n) = P(A)$.

□

4.1 Problems

Problem 1.

We first translate the notation in the book to something more familiar first:

$$\begin{aligned} \limsup_n x_n &= \inf_{n \in \mathbb{N}} \sup_{k \geq n} x_k \\ \liminf_n x_n &= \sup_{n \in \mathbb{N}} \inf_{k \geq n} x_k \end{aligned}$$

Suppose that $\omega \in \limsup_n A_n := A$. Then, $I_A(\omega) = 1$. Also, ω is in infinitely many of the A_n 's. That is, $I_{A_n}(\omega) = 1$ for infinitely many $n \in \mathbb{N}$. In other words, for every $n \in \mathbb{N}$, there exists $k \geq n$ so that $I_{A_k}(\omega) = 1$. That is, $\sup_{k \geq n} I_{A_k}(\omega) = 1$ since the range of I_{A_k} is $\{0, 1\}$. Therefore, $\limsup_n I_{A_n}(\omega) = 1$. Now, suppose that $I_A(\omega) = 0$. Then, for some $n \in \mathbb{N}$, $\omega \notin A_k$ for every $k \geq n$. That is, $I_{A_k}(\omega) = 0$. Then, $\inf_{n \in \mathbb{N}} \sup_{k \geq n} I_{A_k}(\omega) = 0$, implying that $\limsup_n I_{A_n}(\omega) = 0$. This shows that $I_A = I_{\limsup_n A_n} = \limsup_n I_{A_n}$. We now similarly prove that $I_A = \liminf_n I_{A_n}$ where $A = \liminf_n A_n$. Suppose $\omega \in A$. Then, $I_A(\omega) = 1$. Also, $\omega \in A_n$ for all but finitely many $n \in \mathbb{N}$. Then, for all but finitely many $n \in \mathbb{N}$, $I_{A_n}(\omega) = 1$. That is, for every $n \in \mathbb{N}$, there exists $k \in \mathbb{N}$ so that $I_{A_k}(\omega) = 1$ for all $k \geq m \geq n$. Consequently, for any particular $n \in \mathbb{N}$, $\inf_{k \geq n} I_{A_k}(\omega) = 1$. Then, $\liminf_n I_{A_n}(\omega) = 1$. Now suppose $\omega \notin A$. Then, $I_A(\omega) = 0$. There exists $N \in \mathbb{N}$ so that $\omega \notin A_n$ for all $n \geq N$. That is, $I_{A_n}(\omega) = 0$ for all $n \geq N$. Consider any fixed $n \in \mathbb{N}$. If $n \geq N$,

then $\inf_{k \geq n} I_{A_k}(\omega) = 0$ because $I_{A_k}(\omega) = 0$ for all $k \geq n \geq N$. If $n < N$, then still $\inf_{k \geq n} I_{A_k}(\omega) = 0$. So, $I_{A_k}(\omega) = 0$ for all $k \geq N > n$. Therefore, $\liminf_n I_{A_n}(\omega) = 0$. So, $I_A = \liminf_n I_{A_n}$.

Suppose $\limsup_n A_n = \liminf_n A_n$. Let ω be arbitrary. We show $\limsup_n I_{A_n}(\omega) = \liminf_n I_{A_n}(\omega)$. Based on what we just prove, this is equivalent to showing that $I_{\limsup_n A_n}(\omega) = I_{\liminf_n A_n}(\omega)$. Since $\limsup_n A_n = \liminf_n A_n$, this is true, and $\limsup_n I_{A_n}(\omega) = \liminf_n I_{A_n}(\omega)$ for every ω , so $\lim_n I_{A_n}(\omega)$ exists for every ω . Now, suppose $\lim_n I_{A_n}(\omega)$ exists for every ω . Then, $\limsup_n I_{A_n}(\omega) = \liminf_n I_{A_n}(\omega)$ for every ω . By what we just prove, this is equivalent to saying $I_{\limsup_n A_n}(\omega) = I_{\liminf_n A_n}(\omega)$. This would mean that $\omega \in \limsup_n A_n$ if and only if $\omega \in \liminf_n A_n$. Then, $\limsup_n A_n \subset \liminf_n A_n$, so $\limsup_n A_n = \liminf_n A_n$.

Problem 2.

(a)

$$\begin{aligned} \limsup_n (A_n \cap B_n) &= \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} (A_k \cap B_k) \\ &\subset \bigcap_{n=1}^{\infty} \left(\bigcup_{k=n}^{\infty} A_k \cap \bigcup_{k=n}^{\infty} B_k \right) \\ &= \left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k \right) \cap \left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} B_k \right) \\ &= (\limsup_n A_n) \cap (\limsup_n B_n) \end{aligned}$$

$$\begin{aligned} (\limsup_n A_n) \cup (\limsup_n B_n) &= \left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k \right) \cup \left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} B_k \right) \\ &= \bigcap_{n=1}^{\infty} \left(\bigcup_{k=n}^{\infty} A_k \cup \bigcup_{k=n}^{\infty} B_k \right) \\ &= \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} (A_k \cup B_k) \\ &= \limsup_n (A_n \cup B_n) \end{aligned}$$

$$\begin{aligned}
(\liminf_n A_n) \cap (\liminf_n B_n) &= \left(\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k \right) \cap \left(\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} B_k \right) \\
&= \bigcup_{n=1}^{\infty} \left(\bigcap_{k=n}^{\infty} A_k \cap \bigcap_{k=n}^{\infty} B_k \right) \\
&= \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} (A_k \cap B_k) \\
&= \liminf_n (A_n \cap B_n)
\end{aligned}$$

$$\begin{aligned}
(\liminf_n A_n) \cup (\liminf_n B_n) &= \left(\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k \right) \cup \left(\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} B_k \right) \\
&\subset \bigcup_{n=1}^{\infty} \left(\bigcap_{k=n}^{\infty} A_k \cup \bigcap_{k=n}^{\infty} B_k \right) \\
&\subset \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} (A_k \cup B_k) \\
&= \liminf_n (A_n \cup B_n)
\end{aligned}$$

For the examples: Let $A_n = \{x\}$ if n is even and $A_n = \emptyset$ when n is odd. Let $B_n = \emptyset$ when n is even and $B_n = \{x\}$ when n is odd. Then, for each $n \in \mathbb{N}$, $A_n \cap B_n = \emptyset$. So, $\limsup_n (A_n \cap B_n) = \emptyset$. However, $\limsup_n A_n = \limsup_n B_n = \{x\}$. So, $(\limsup_n A_n) \cap (\limsup_n B_n) = \{x\}$, which properly contains $\emptyset = \limsup_n (A_n \cap B_n)$. Also, $\liminf_n (A_n \cup B_n) = \{x\}$. However, $\liminf_n A_n = \liminf_n B_n = \emptyset$. So, $(\liminf_n A_n) \cup (\liminf_n B_n) = \emptyset \cup \emptyset = \emptyset \subset \liminf_n (A_n \cup B_n) = \{x\}$.

(b) TODO

(c)

$$\begin{aligned}
(\liminf_n A_n)^c &= \left(\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k \right)^c = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k^c = \limsup_n A_n^c \\
(\limsup_n A_n)^c &= \left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k \right)^c = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k^c = \liminf_n A_n^c
\end{aligned}$$

$$\begin{aligned}
\limsup_n (A_n \cap A_{n+1}^c) &= \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} (A_k \cap A_{k+1}^c) \\
&= \bigcap_{n=1}^{\infty} \left[\left(\bigcup_{k=n}^{\infty} A_k \right) \cap \left(\bigcup_{k=n}^{\infty} A_{k+1}^c \right) \right] \\
&= \bigcap_{n=1}^{\infty} \left[\left(\bigcup_{k=n}^{\infty} A_k \right) \cap \left(\bigcup_{k=n+1}^{\infty} A_k^c \right) \right] \\
&= \left[\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k \right] \cap \left[\bigcap_{n=1}^{\infty} \bigcup_{k=n+1}^{\infty} A_k^c \right] \\
&= \left[\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k \right] \cap \left[\bigcap_{n=1}^{\infty} \left(\left\{ \bigcup_{k=n}^{\infty} A_k^c \right\} \setminus A_n^c \right) \right] \\
&= \left[\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k \right] \cap \left[\bigcap_{n=1}^{\infty} \left(\left(\bigcup_{k=n}^{\infty} A_k^c \right) \cap A_n \right) \right] \\
&= \left[\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k \right] \cap \left[\bigcap_{n=1}^{\infty} \left(\bigcup_{k=n}^{\infty} A_k^c \right) \right] \cap \bigcap_{n=1}^{\infty} A_n \\
&= \left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k \right) \cap \left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k^c \right) \\
&= \left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k \right) \setminus \left(\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k \right) \\
&= \limsup_n A_n - \liminf_n A_n
\end{aligned}$$

$$\begin{aligned}
\limsup_n (A_n^c \cap A_{n+1}) &= \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} (A_k^c \cap A_{k+1}) \\
&= \bigcap_{n=1}^{\infty} \left[\left(\bigcup_{k=n}^{\infty} A_k^c \right) \cap \left(\bigcup_{k=n}^{\infty} A_{k+1} \right) \right] \\
&= \bigcap_{n=1}^{\infty} \left[\left(\bigcup_{k=n}^{\infty} A_k^c \right) \cap \left(\bigcup_{k=n+1}^{\infty} A_k \right) \right] \\
&= \left[\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k^c \right] \cap \left[\bigcap_{n=1}^{\infty} \bigcup_{k=n+1}^{\infty} A_k \right] \\
&= \left[\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k^c \right] \cap \left[\bigcap_{n=1}^{\infty} \left\{ \left(\bigcup_{k=n}^{\infty} A_k \right) \setminus A_n \right\} \right] \\
&= \left[\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k^c \right] \cap \left[\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k \right] \cap \bigcap_{n=1}^{\infty} A_n^c \\
&= \left[\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k^c \right] \cap \left[\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k \right] \\
&= \left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k \right) \setminus \left(\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k \right) \\
&= \limsup_n A_n - \liminf_n A_n
\end{aligned}$$

- (d) If $A_n \rightarrow A$, then $\limsup_n A_n = \liminf_n A_n = A$. If $B_n \rightarrow B$, then $\limsup_n B_n = \liminf_n B_n = B$. We know from part (a) that $\limsup_n (A_n \cup B_n) = (\limsup_n A_n) \cup (\limsup_n B_n) = A \cup B$. However, this also means that $\limsup_n (A_n \cup B_n) = (\limsup_n A_n) \cup (\limsup_n B_n) = (\liminf_n A_n) \cup (\liminf_n B_n) \subset \liminf_n (A_n \cup B_n)$. Since we have shown that $\limsup_n (A_n \cup B_n) \subset \liminf_n (A_n \cup B_n)$, it follows that $A \cup B = \limsup_n (A_n \cup B_n) = \liminf_n (A_n \cup B_n)$. Hence, if $A_n \rightarrow A$ and $B_n \rightarrow B$, then $A_n \cup B_n \rightarrow A \cup B$. To prove the case for intersections, we show that: if $D_n \rightarrow D$, then $D_n^c \rightarrow D^c$ for some sequence of sets $\{D_n\}_{n \rightarrow \mathbb{N}}$. Since $\limsup_n D_n = \liminf_n D_n = D$, it follows that $(\limsup_n D_n)^c = (\liminf_n D_n)^c = D^c$. By part (a), we know this means $\liminf_n D_n^c = \limsup_n D_n^c = D^c$. Therefore, $D_n^c \rightarrow D^c$. Applying this fact, $A_n \rightarrow A$ implies $A_n^c \rightarrow A^c$ and $B_n \rightarrow B$ implies $B_n^c \rightarrow B^c$. Using what we just prove about unions, $A_n^c \cup B_n^c \rightarrow A^c \cup B^c$. Again, applying the fact for complements, we get that $A_n \cap B_n \rightarrow A \cap B$.